Hints: Integrals – 1. Substitutions: rationalisation, Euler, the half-angle substitution

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. For $\int \sqrt{\tan x}\,dx$ put $t=\sqrt{\tan x}$, i.e. $x=\arctan t^2$, $dx=\frac{2t\,dt}{1+t^4}$; you get $\int \frac{2t^2}{1+t^4}dt$. For the second integral divide numerator and denominator by $\cos^4 x$: with $t=\tan x$ it becomes $\int \frac{(1+t^2)\,dt}{1+t^4}$.

Hint 2. The trick for $\frac{1\pm t^2}{1+t^4}$: divide by t^2 to get $\frac{1\pm 1/t^2}{t^2+1/t^2}$, and note $t^2+\frac{1}{t^2}=(t-\frac{1}{t})^2+2=(t+\frac{1}{t})^2-2$. So $u=t-\frac{1}{t}$ or $u=t+\frac{1}{t}$ reduces everything to $\int \frac{du}{u^2\pm 2}$, a quadratic. Write $2t^2=(t^2+1)+(t^2-1)$ for the first integral.

Hint 3. Answers: $\int \frac{dx}{\sin^4 x+\cos^4 x}=\frac{1}{\sqrt{2}}\arctan\frac{\tan^2 x-1}{\sqrt{2}\tan x}+C$ (equivalently $\frac{1}{\sqrt{2}}\arctan\frac{\tan 2x}{\sqrt{2}}$, up to constants on intervals). With $t=\sqrt{\tan x}$: $\int \sqrt{\tan x}\,dx=\frac{1}{\sqrt{2}}\arctan\frac{t-1/t}{\sqrt{2}}+\frac{1}{2\sqrt{2}}\ln\frac{t+1/t-\sqrt{2}}{t+1/t+\sqrt{2}}+C$. With $s=\tan\frac{x}{2}$ instead, $\sin^4 x+\cos^4 x=\frac{16s^4+(1-s^2)^4}{(1+s^2)^4}$, a quartic in s^2 in the denominator: much worse.

Problem 2

Hint 1. If $P(x)=\prod(x-\omega_k)$ has simple roots, then $\frac{1}{P(x)}=\sum_k \frac{1}{P'(\omega_k)(x-\omega_k)}$. For $P=x^n+1$: $P'(\omega)=n\omega^{n-1}=-n/\omega$, so $\frac{1}{x^n+1}=-\frac{1}{n}\sum_k\frac{\omega_k}{x-\omega_k}$; for x^6-1 similarly $\frac{1}{x^6-1}=\frac{1}{6}\sum\frac{\zeta_k}{x-\zeta_k}$.

Hint 2. Pair conjugate roots $e^{\pm i\theta}$: $\frac{e^{i\theta}}{x-e^{i\theta}}+\frac{e^{-i\theta}}{x-e^{-i\theta}}=\frac{2x\cos\theta-2}{x^2-2x\cos\theta+1}$, which integrates to a log plus an arctan. This is the general answer for $\int\frac{dx}{x^n+1}$, with $\theta=\frac{(2k+1)\pi}{n}$ (plus the real root -1 when n is odd).

Hint 3. $\int\frac{dx}{x^4+1}=\frac{1}{4\sqrt{2}}\ln\frac{x^2+\sqrt{2}x+1}{x^2-\sqrt{2}x+1}+\frac{1}{2\sqrt{2}}\Big(\arctan(\sqrt{2}x+1)+\arctan(\sqrt{2}x-1)\Big)+C$. For x^6-1: $\frac{1}{x^6-1}=\frac{1}{6}\Big(\frac{1}{x-1}-\frac{1}{x+1}+\frac{x-2}{x^2-x+1}-\frac{x+2}{x^2+x+1}\Big)$, giving $\frac{1}{6}\ln\left|\frac{x-1}{x+1}\right|+\frac{1}{12}\ln\frac{x^2-x+1}{x^2+x+1}-\frac{1}{2\sqrt{3}}\Big(\arctan\frac{2x-1}{\sqrt{3}}+\arctan\frac{2x+1}{\sqrt{3}}\Big)+C$.

Problem 3

Hint 1.

(a) $u=2+4x$; (b) multiply by $\frac{\sec x+\tan x}{\sec x+\tan x}$; (c) the numerator is half the derivative of the denominator; (d) $\frac{1}{(x+1/x)^2}=\frac{x^2}{(x^2+1)^2}$; (e) divide: $\frac{x^3}{1+x^2}=x-\frac{x}{1+x^2}$; (f) factor numerator and denominator by grouping.

Hint 2. (f): $x^5-x^3+x^2-1=(x^2-1)(x^3+1)$ and $x^4-x^3+x-1=(x-1)(x^3+1)$, so the integrand is $x+1$. (d): $\frac{x^2}{(x^2+1)^2}=\frac{1}{x^2+1}-\frac{1}{(x^2+1)^2}$, or integrate by parts with $u=x$.

Hint 3. Answers $(+C)$: (a) $\frac{1}{40}(2+4x)^{5/2}-\frac{1}{12}(2+4x)^{3/2}$; (b) $\ln|\sec x+\tan x|$; (c) $\frac{1}{2}\ln(x^2+2x+3)$; (d) $\frac{1}{2}\Big(\arctan x-\frac{x}{x^2+1}\Big)$; (e) $\frac{x^2}{2}-\frac{1}{2}\ln(1+x^2)$; (f) $\frac{x^2}{2}+x$.

Problem 4

Hint 1. $\sqrt{1+\tan^2 x} = \sqrt{\sec^2 x} = \frac{1}{|\cos x|}$, not $\sec x$. So the integrand is $\frac{\sin x}{|\cos x|} = \pm\tan x$ depending on the sign of $\cos x$.

Hint 2. On intervals where $\cos x > 0$ the antiderivative is $-\ln|\cos x| + C$; where $\cos x < 0$ it is $+\ln|\cos x| + C$. In one formula: $-\operatorname{sgn}(\cos x)\ln|\cos x| + C$ on each interval. The naive answer $-\ln|\cos x|$ silently used $\sqrt{\sec^2 x} = \sec x$, true only where $\cos x > 0$.

Problem 5

Hint 1. For I_n, integrate $I_{n-1} = \int \frac{dx}{(x^2+a^2)^{n-1}}$ by parts with $u = (x^2+a^2)^{-(n-1)}$, $dv = dx$, and write $x^2 = (x^2+a^2) - a^2$ in the new integral. For J_n, write $\sin^n x = \sin^{n-1} x \cdot \sin x$, take $dv = \sin x\,dx$, and use $\cos^2 = 1 - \sin^2$.

Hint 2. You should get $I_n = \frac{x}{2(n-1)a^2(x^2+a^2)^{n-1}} + \frac{2n-3}{2(n-1)a^2} I_{n-1}$ with $I_1 = \frac{1}{a}\arctan\frac{x}{a}$, and $J_n = -\frac{\sin^{n-1} x\cos x}{n} + \frac{n-1}{n} J_{n-2}$ with $J_0 = x$.

Hint 3. $I_3 = \frac{x}{4a^2(x^2+a^2)^2} + \frac{3x}{8a^4(x^2+a^2)} + \frac{3}{8a^5}\arctan\frac{x}{a} + C$ and $J_6 = -\frac{\sin^5 x\cos x}{6} - \frac{5\sin^3 x\cos x}{24} - \frac{5\sin x\cos x}{16} + \frac{5x}{16} + C$.

Problem 6

Hint 1. Make the inverse function (or the inner root) disappear by differentiating it: (b), (c), (d), (e) take $u =$ the inverse function and $dv = dx$ (or $x\,dx$). In (a) substitute $t = \sqrt{x}$ first; in (f) substitute $t = e^x$ first.

Hint 2.

(a) becomes $2\int t^2 e^t\,dt$; (f) becomes $\int t^2 \arctan t\,dt$, then parts with $u = \arctan t$ and divide $\frac{t^3}{1+t^2} = t - \frac{t}{1+t^2}$. (c) needs parts twice: the second time with $u = \arcsin x$, $dv = \frac{x\,dx}{\sqrt{1-x^2}}$. In (d) the leftover is $\int \frac{\sqrt{x}}{2(1+x)}dx$, done by $x = t^2$.

Hint 3. Answers $(+C)$: (a) $2e^{\sqrt{x}}(x - 2\sqrt{x} + 2)$; (b) $\frac{x^2+1}{2}\arctan x - \frac{x}{2}$; (c) $x\arcsin^2 x + 2\sqrt{1-x^2}\arcsin x - 2x$; (d) $(x+1)\arctan\sqrt{x} - \sqrt{x}$; (e) $x\operatorname{arcosh} x - \sqrt{x^2-1}$; (f) $\frac{e^{3x}}{3}\arctan e^x - \frac{e^{2x}}{6} + \frac{1}{6}\ln(1+e^{2x})$.

Problem 7

Hint 1. Rule: if $R(-\sin x, \cos x) = -R(\sin x, \cos x)$ use $t = \cos x$; if $R(\sin x, -\cos x) = -R$ use $t = \sin x$; if $R(-\sin x, -\cos x) = R$ use $t = \tan x$. (Reason: in the first case $R\,dx = R_1(\cos x, \sin^2 x)\sin x\,dx$, etc.)

Hint 2. $\frac{\sin^3 x}{1+\cos^2 x}$ is odd in sin: $t = \cos x$ gives $\int \frac{t^2-1}{t^2+1}dt$, so $\cos x - 2\arctan(\cos x) + C$. $\frac{1}{1+\cos^2 x}$ is even: $t = \tan x$ gives $\int \frac{dt}{t^2+2} = \frac{1}{\sqrt{2}}\arctan\frac{\tan x}{\sqrt{2}} + C$. $\frac{\cos x}{2-\cos^2 x} = \frac{\cos x}{1+\sin^2 x}$ is odd in cos: $t = \sin x$ gives $\arctan(\sin x) + C$.

Problem 8

Hint 1. Complete the square and use a trigonometric or hyperbolic substitution: $x - x^2 = \frac{1}{4} - (x - \frac{1}{2})^2$; $12 - 3x^2 = 3(4 - x^2)$ with $x = 2\sin\theta$; $\sqrt{x^2-1}$ with $x = \sec\theta$ (or $\cosh u$); $\sqrt{x^2+25}$ with $x = 5\sinh u$ (or $5\tan\theta$).

Hint 2. Answers $(+C)$: $\arcsin(2x-1)$ (equivalently $2\arcsin\sqrt{x}$); $\sqrt{3}\left(\frac{x}{2}\sqrt{4-x^2} + 2\arcsin\frac{x}{2}\right)$; for $x > 1$, $\int \tan^2\theta\,d\theta$ gives $\sqrt{x^2-1} - \operatorname{arcsec} x$; $\frac{x}{2}\sqrt{x^2+25} + \frac{25}{2}\ln\left(x + \sqrt{x^2+25}\right)$.

Problem 9

Hint 1. By parts twice, $I = \int e^{ax}\cos bx\,dx$ reappears on the right with coefficient $-\frac{b^2}{a^2}$; solve the resulting linear equation for I. Alternatively $\int e^{(a+ib)x}dx = \frac{e^{(a+ib)x}}{a+ib}$; multiply by $\frac{a-ib}{a^2+b^2}$ and take the real part.

Hint 2. Both give $\frac{e^{ax}(a\cos bx+b\sin bx)}{a^2+b^2}+C$, and the imaginary part gives $\int e^{ax}\sin bx\,dx$ for free; the two integrations by parts are exactly the real and imaginary parts of the single complex computation.

Problem 10

Hint 1. A denominator v^2 suggests $(u/v)' = \frac{u'v-uv'}{v^2}$; a fraction whose numerator is the derivative of its denominator integrates to $\ln|v|$. In (a) and (e) try $v = \log x$, $u = \sin x$. In (b) divide numerator and denominator by x.

Hint 2.

(b) $\frac{x+1}{x(x+\log x)} = \frac{1+1/x}{x+\log x}$; (c) $1+\cos x = (x+\sin x)'$. (d) is not quite of this form: write $1+2x^{2022} = (1+x^{2022})+x^{2022}$, so the integrand is $\frac{1}{x}+\frac{x^{2021}}{1+x^{2022}}$. (e) equals $\frac{\cos x}{\log x}-\frac{\sin x}{x\log^2 x}$, the same as (a).

Hint 3. Answers ($+C$): (a) and (e) $\frac{\sin x}{\log x}$; (b) $\ln|x+\log x|$; (c) $\ln|x+\sin x|$; (d) $\ln|x|+\frac{1}{2022}\ln(1+x^{2022})$.